Custom Reinforcement Learning Agent Implementation using Farama Gymnasium

Course: Topics in AI | Assignment No. I/CLO-I

1. Problem Description

The selected environment is FrozenLake-v1 from Gymnasium with the slippery setting disabled. The agent observes a discrete state index representing the current tile in a 4x4 grid. The action space contains four possible moves: left, down, right, and up. The reward structure is sparse: the agent receives 1.0 upon reaching the goal tile and 0.0 elsewhere, with falling into a hole causing termination. This environment is a classic small-scale control problem that requires learning a policy that avoids unsafe tiles while navigating to the reward state.

2. Methodology

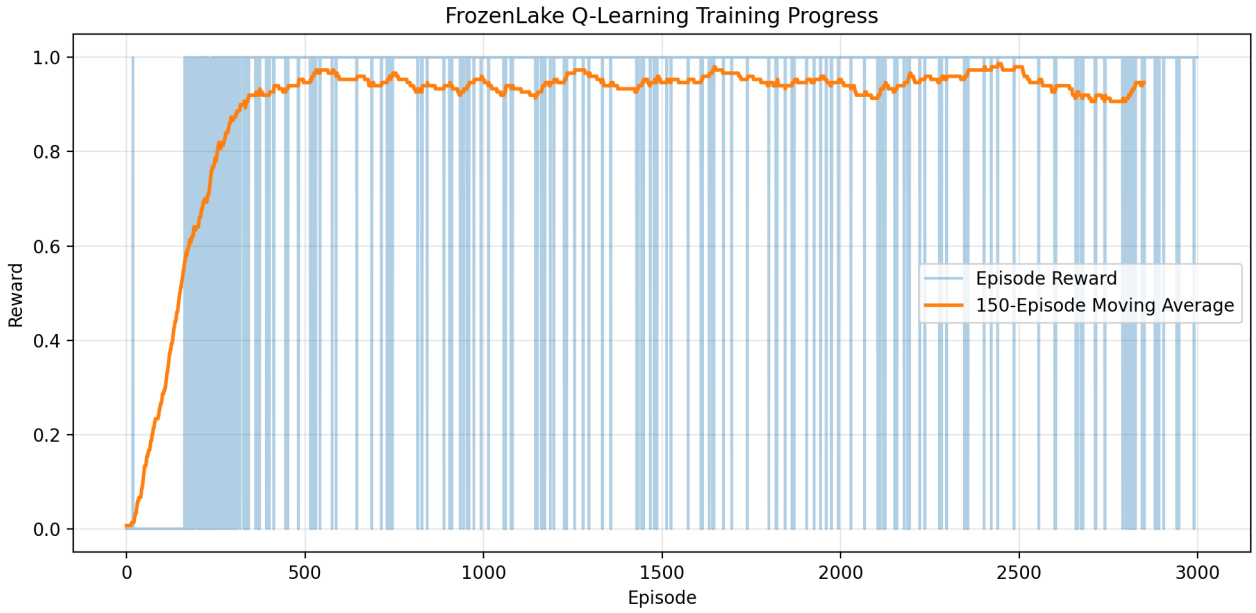
The implemented method is tabular Q-learning, a model-free RL algorithm that estimates the action-value function Q(s, a). The update rule used is: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$, where α is the learning rate, γ is the discount factor, and r is the immediate reward. The agent follows an ε-greedy policy during training, balancing exploration and exploitation. Once the Q-table converges, the greedy policy $\arg\max_a Q(s, a)$ is used for evaluation and policy execution.

Experimental Setup & Hyperparameters

Parameter	Value
States	16
Actions	4
Learning Rate (α)	0.8
Discount Factor (γ)	0.99
Initial ε	1.0
Minimum ε	0.05
ε Decay	0.995
Episodes	3000
Max Steps per Episode	200

3. Results & Discussion

The training curve shows rapid learning in the early episodes, followed by stable convergence near the optimal policy. The agent reaches a perfect evaluation score over 200 test episodes, yielding an average return of 1.0 and a success rate of 100%. The learned behavior avoids holes and consistently reaches the goal tile. The main observed challenge was exploration in the early phase, which is why the agent starts with a relatively high ε and decays it gradually. This encourages sufficient exploration before exploitation takes over.



4. Conclusion

The Q-learning implementation successfully solved the FrozenLake environment. The trained policy demonstrates stable convergence, robust performance, and a clear interpretation of the learned state-action values. This assignment confirms the effectiveness of tabular RL methods for small discrete environments and provides a reproducible baseline for further experimentation with deeper algorithms or more complex environments.

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